FocusFlow - Design Document

Project Description

FocusFlow is a productivity application that helps users stay focused and organized. It combines task management with the Pomodoro technique, allowing users to create and prioritize tasks, run timed focus sessions, and track their productivity over time. Users can view session history, monitor streaks, and see visual stats on their work habits, helping them build consistency and get more done.

Tech Stack

- **Frontend:** React 18 with Hooks, React Router, Vite
- **Backend:** Node.js, Express 5
- **Database:** MongoDB (native driver, no Mongoose)
- **Authentication:** JWT, bcryptjs

User Personas

Persona 1: Alex

- **Age:** 22
- **Occupation:** Computer Science Student
- **Background:** Alex is a college student who often struggles with procrastination. He finds it hard to stay focused during long study sessions and frequently loses track of how much time he's actually spending on assignments.
- **Goals:**
  - Build better study habits
  - Track productivity over time
  - Stay motivated through visible progress
- **Frustrations:**
  - Gets distracted easily
  - Doesn't know how much time tasks actually take
  - Loses motivation without seeing progress
- **Needs:** Structured focus sessions with timers and the ability to track study habits over time.

Persona 2: Maya

- **Age:** 28
- **Occupation:** Freelance Designer
- **Background:** Maya juggles multiple client projects simultaneously. She needs to track billable hours and ensure she's making progress on all her commitments. Without organization, she risks missing deadlines.
- **Goals:**
    - Organize tasks by client and priority
    - Track time spent on each project
    - Visualize productivity patterns
- **Frustrations:**
    - Loses track of tasks across multiple projects
    - Doesn't know how long she spends on each client
    - No clear picture of daily/weekly productivity
- **Needs:** Task management with categories and priorities, plus time tracking and productivity dashboards.

User Stories

Authentication & Sessions (Nishant)

ID	User Story	Acceptance Criteria
1	As Alex, I want to register, log in, and log out so my data is saved securely across devices.	User can create account with email/password, log in, and log out. JWT token stored for session persistence.
2	As Alex, I want to start, pause, and reset a Pomodoro timer so I can focus without distractions.	Timer displays countdown, Start/Pause/Reset buttons work correctly, audio notification on completion.
3	As Alex, I want to customize my timer duration so I can adjust sessions to my preference.	User can set custom duration (1-60 minutes) before starting timer.
4	As Alex, I want each completed session saved automatically so I have a record of my work.	Completed sessions are saved to database with duration, label, type, and timestamp.
5	As Alex, I want to view my session history and streaks so I can stay motivated.	Session history displays all past sessions, streak counter shows consecutive days, weekly graph visualizes activity.

Task Management (Avijit/Vee)

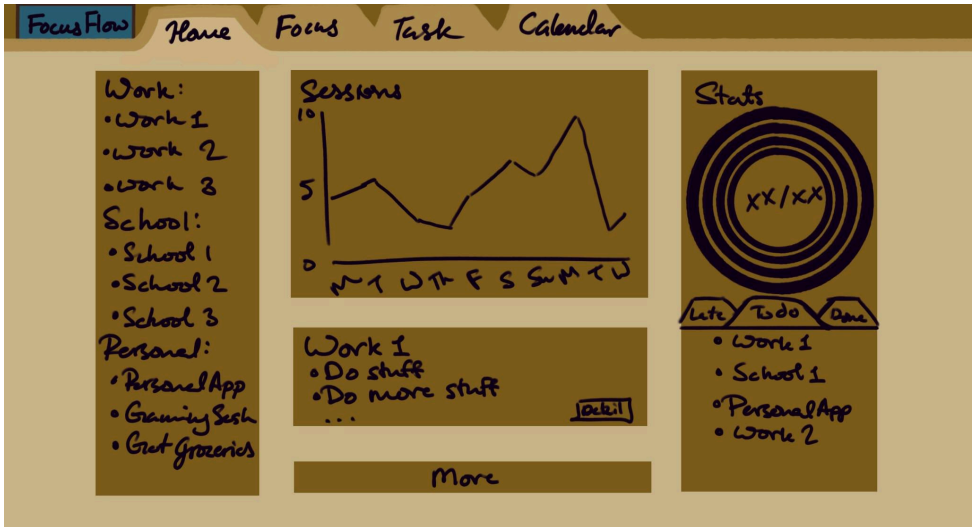
ID	User Story	Acceptance Criteria
6	As Maya, I want to create, edit, and delete tasks so I can manage my workload.	Full CRUD operations on tasks with title, description, due date.
7	As Maya, I want to mark tasks as complete so I can track my progress.	Toggle task completion status, visual indicator for completed tasks.
8	As Maya, I want to set priority levels and categories so I can organize my work.	Tasks have priority (Low/Medium/High) and category fields.
9	As Maya, I want to filter and sort tasks by category, priority, or status so I can find what I need.	Filter dropdowns and sort options on task list view.
10	As Maya, I want a dashboard showing tasks completed and focus stats so I can measure my productivity.	Dashboard displays task completion rate, focus minutes, and productivity charts.

Design Mockups

Home Page

The home page displays a three-column layout:

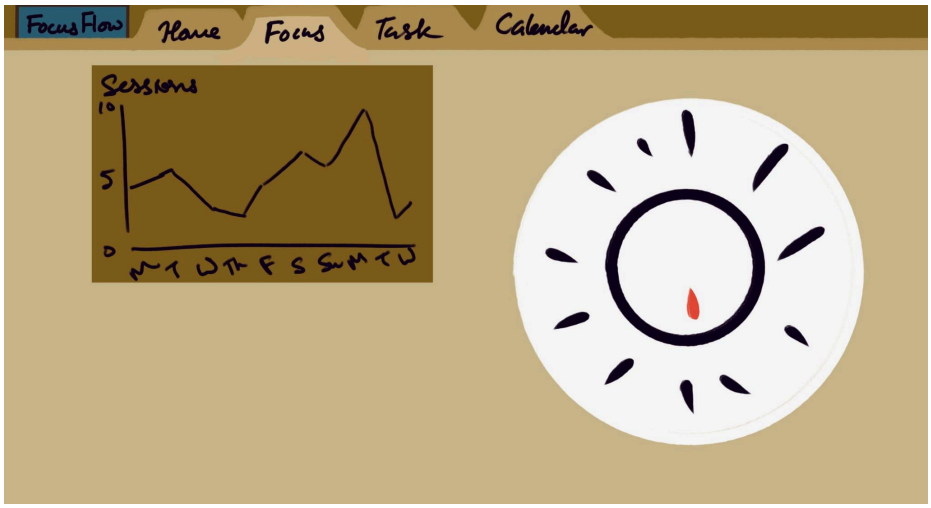
- **Left:** Task overview organized by category (Work, School, Personal)
- **Center:** Sessions graph showing weekly activity, current task details, and "More" button
- **Right:** Stats circle with date, Late/TODO/Done tabs, and task list



Focus Page

The focus page features:

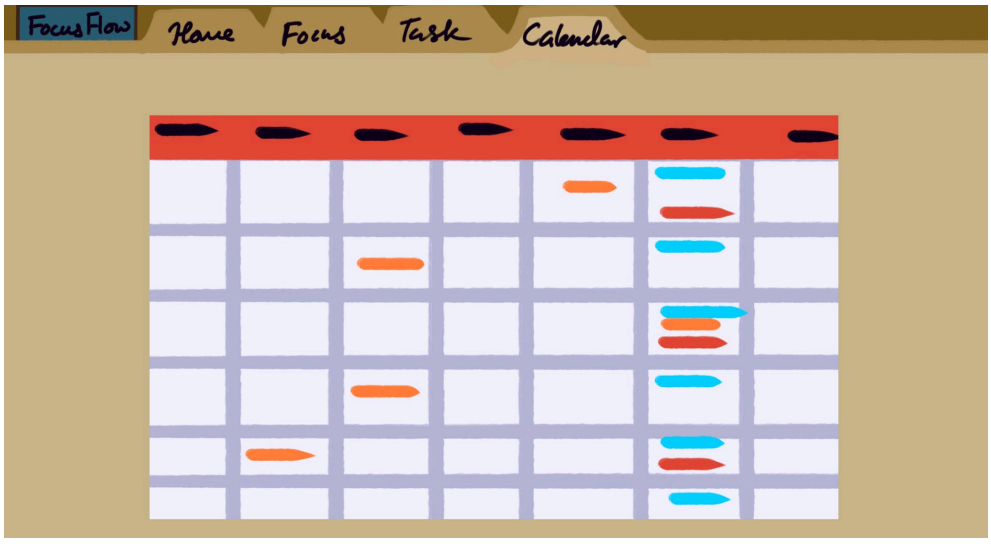
- **Left:** Sessions line graph showing productivity over the week
- **Right:** Large circular timer with visual indicators



Calendar Page

The calendar page shows:

- Monthly calendar grid view
- Color-coded events/tasks (orange, red, blue indicators)
- Days of the week header



Authors

- **Nishant Naravarajula** - Authentication, Timer, Sessions, Session History, Streaks
- **Avijit Singh (Vee)** - Task Management, Calendar, Dashboard

Course

CS 5610 - Web Development
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